



Health Care

University of Missouri – University Hospital

Cumulative Antimicrobial Susceptibility Report

January – December 2025

Microbiology Lab	573-882-1287
Infection Prevention	573-882-6880

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Legend	
80-100%	Good susceptibility
41-79%	Moderate susceptibility
0-40%	Poor susceptibility

Gram Positive Organisms % Susceptible
University of Missouri – University Hospital
All locations and ages (excluding clinics)
January 1-December 31, 2025

	# of Isolates	Penicillin (IV)	Oxacillin	Ampicillin	Cefotaxime/ ceftriaxone	Clindamycin	Erythromycin	Vancomycin	Rifampin^	Linezolid	Tetracycline	Sulfamethoxazole/ trimethoprim	Levofloxacin	Gentamicin synergy	Nitrofurantoin	# Isolates tested with nitrofurantoin
<i>Staphylococcus aureus</i> (MSSA)	491	-	100	-	-	95	-	100	-	-	94	98	-	-	100	25
<i>Staphylococcus aureus</i> (MRSA)	335					70	-	100	99	-	81	73	-	-	100	22
<i>Staphylococcus coagulase negative</i> (CONS)	568	-	38	-	-	64	39	100	-	-	78	-	71	-	100	54
<i>Streptococcus agalactiae</i> (GBS)	58	100	-	-	-	51	-	100	-	-	-	-	-	-	-	-
<i>Streptococcus anginosus</i>	48	98	-	-	-	65	47	100	-	-	-	-	-	-	-	-
<i>Streptococcus pneumoniae</i> (non-sterile)	58	100	-	-	-	-	51	-	-	-	-	81	100	-	-	-
<i>Streptococcus pneumoniae</i> (blood/cerebrospinal fluid)*	13	77	-	-	75	-	-	-	-	-	-	-	-	-	-	-
<i>Streptococcus pyogenes</i> (GAS)	36	100	-	-	-	88	80	100	-	-	-	-	-	-	-	-
<i>Enterococcus</i> spp. (all species)	389	-	-	77			-	79	-	96	9		-	76	84	234
<i>Enterococcus faecalis</i>	277	-	-	100			-	99	-	-	-		-	75	99	161
<i>Enterococcus faecium</i>	61	-	-	19			-	55	-	100	-		-	-	28	18

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Susceptibility is reported based on the number of isolates (at minimum 30 isolates) tested against a specific antimicrobial agent. Data for fewer than 30 isolates per organism, per selected time frame is not statistically significant according to CLSI standards. Please exercise discretion when data are reviewed for species with fewer than 30 isolates. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalixin.

(-) Drug was not tested, drug was not tested on >= 30 isolates, or drug is not indicated.

^ Rifampin is 100% susceptible against all *Staphylococcus aureus* strains. Use only as adjunct to main therapy.

* Total number of *Streptococcus pneumoniae* isolates identified in a cerebrospinal fluid culture was one. The remaining organisms were isolated from blood cultures. Not all susceptibility results include the entire specimen count of 14. Interpret results with caution. Susceptibility results are representative of the meningitis breakpoint. See page 8 for *S. pneumoniae* breakpoint details.

Gram Negative Organisms % Susceptible
University of Missouri – University Hospital
All locations and ages (excluding clinics)
January 1-December 31, 2025

	# of Isolates	Ampicillin	Ampicillin/sulbactam	Piperacillin/tazobactam	Cefazolin ^s (urinary isolates only)	Cefoxitin	Cefotaxime/ceftriaxone	Ceftazidime	Cefepime	Meropenem	Amikacin [^]	Gentamicin	Tobramycin	Minocycline	Sulfamethoxazole/ trimethoprim	Ciprofloxacin	Levofloxacin	Nitrofurantoin (urinary isolates only)	# Isolates tested with nitrofurantoin
<i>Citrobacter freundii</i> *	47			-			-	-	98	98	-	91	-	-	-	77	-	100	35
<i>Enterobacter cloacae</i> *	135			-			-	-	93	100	-	97	87	-	92	94	91	39	57
<i>Escherichia coli</i>	1265	52	65	96	95	83	92	94	96	100	-	90	91	-	75	74	70	97	1050
<i>Klebsiella aerogenes</i> *	47			-			-	-	98	100	-	100	-	-	100	91	-	10	21
<i>Klebsiella oxytoca</i>	57		64	-	-	91	94	-	98	100	-	93	-	-	89	93	88	86	21
<i>Klebsiella pneumoniae</i>	322		73	88	100	87	81	84	90	98	-	93	94	-	79	77	76	28	228
<i>Proteus mirabilis</i>	176	72	82	99	94	93	89	98	93	99	-	87	91	-	62	57	71		
<i>Pseudomonas aeruginosa</i>	304			94				90	94	90	100	-	-			85	81	-	-
<i>Raoultella ornithinolytica</i>	49		68	-	-	-	88	-	98	100	-	98	-	-	90	96	-	85	26
<i>Serratia marcescens</i>	62			-		-	92	-	97	100	-	98	-	-	-	95	94		
<i>Stenotrophomonas maltophilia</i>	64							-	-					91	91	-	91	-	-

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Susceptibility is reported based on the number of isolates (at minimum 30 isolates) tested against a specific antimicrobial agent. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalexin. Data for fewer than 30 isolates per organism, per selected time frame is not statistically significant according to CLSI standards. Please exercise discretion when data are reviewed for species with fewer than 30 isolates.

Gentamicin is no longer recommended for the management of *Pseudomonas* spp. per the Clinical and Laboratory Standards Institute.

(-) Drug was not tested, drug was not tested on ≥ 30 isolates, or drug is not indicated.

■ Organism exhibits intrinsic resistance to this drug.

§ Number of *E. coli* isolates susceptible to cefazolin with a MIC ≤ 2 was 406/930 based on micro reporting. This number may be higher than what is reported.

[^]Amikacin used for *Pseudomonas aeruginosa* should only be used for urinary tract infections.

(*) HECK-Yes bacteria include *Hafnia alvei*, *Enterobacter cloacae* complex, *Citrobacter freundii*, *Klebsiella aerogenes*, and *Yersinia enterocolitica*. These organisms are recognized by the IDSA in the 2024 Guidance on the Treatment of Antimicrobial-Resistant Gram-Negative Infections to carry a risk for the emergence of clinically relevant *ampC* expression during antibiotic treatment. Basal production of AmpC beta-lactamases in these organisms lends to intrinsic resistance to ampicillin, amoxicillin/clavulanate, ampicillin/sulbactam, and first- and second-generation cephalosporins.

Emergency Department (Main)
University of Missouri – University Hospital
All ages (excluding clinics)
January 1, 2024-December 31, 2025

		# of Isolates	Penicillin (IV)	Oxacillin	Ampicillin	Ampicillin/sulbactam	Piperacillin/tazobactam	Cefazolin (urinary isolates only)	Cefoxitin	Cefotaxime/ceftriaxone	Ceftazidime	Cefepime	Meropenem	Gentamicin	Tobramycin	Clindamycin	Erythromycin	Vancomycin	Tetracycline	Sulfamethoxazole/ trimethoprim	Ciprofloxacin	Levofloxacin	Nitrofurantoin (urinary isolates only)	# Isolates tested with nitrofurantoin	
Gram Positive	<i>Staphylococcus aureus</i> (MSSA)	316	-	100	-	-	-	-	-	-	-	-	-	-	-	94	-	100	95	96	-	-	100	36	
	<i>Staphylococcus aureus</i> (MRSA)	294												-	-	69	-	100	84	65	-	-	100	39	
	<i>Staphylococcus coagulase negative</i> (CONS)	466	-	42	-	-	-	-	-	-		-	-	-	-	63	38	100	76	-	-	73	100	54	
	<i>Streptococcus agalactiae</i> (GBS)	47	100	-	-	-	-	-	-	-		-	-	-	-	45	-	100	-	-	-	-	-	-	
	<i>Streptococcus anginosus</i>	49	93	-	-	-	-	-	-	-		-	-	-	-	70	61	100	-	-	-	-	-	-	
	<i>Streptococcus pyogenes</i> (GAS)	54	100	-	-	-	-	-	-	-		-	-	-	-	82	77	100	-	-	-	-	-	-	
	<i>Enterococcus</i> spp. (all species)	298	-	-	87	-	-						-					3	89	-		-	-	89	213
	<i>Enterococcus faecalis</i>	245	-	-	100	-	-						-					-	99	-		-	-	99	176
Gram Negative	<i>Citrobacter freundii</i> *	42					-		-	-	-	100	100	90	-				-	-	76	-	84	38	
	<i>Citrobacter koseri</i>	30				-	-	-	-	-	-	97	-	97	-				-	-	97	-	89	27	
	<i>Enterobacter cloacae</i> *	111					-		-	-	-	92	100	96	91				-	87	92	94	32	80	
	<i>Escherichia coli</i>	1720			51	63	96	96	79	91	96	98	100	91	92				-	74	75	67	96	1556	
	<i>Klebsiella aerogenes</i> *	42					-		-	-	-	100	100	100	-				-	100	98	-	12	34	
	<i>Klebsiella oxytoca</i>	48				79	-	-	-	96	-	100	100	94	-				-	96	94	-	85	26	
	<i>Klebsiella pneumoniae</i>	344				77	91	98	85	84	87	94	100	94	96				-	81	80	70	26	295	
	<i>Proteus mirabilis</i>	211			72	84	99	93	87	90	99	95	100	90	89					72	59	57			
	<i>Pseudomonas aeruginosa</i>	204					95				92	94	91	-	-						86	79	-	-	
	<i>Raoultella</i> spp.	43				78	-	-	-	92	97	97	100	98	-				-	94	93	-	84	38	
	<i>Serratia marcescens</i>	47						-		-	98	-	100	100	98	-				-	94	-			

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Susceptibility is reported based on the number of isolates (at minimum 30 isolates) tested against a specific antimicrobial agent. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalixin. Data for fewer than 30 isolates per organism, per selected time frame is not statistically significant according to CLSI standards. Please exercise discretion when data are reviewed for species with fewer than 30 isolates. Gentamicin is no longer recommended for the management of *Pseudomonas* spp. per the Clinical and Laboratory Standards Institute.

(-) Drug was not tested, drug was not tested on >= 30 isolates, or drug is not indicated.

▪ Organism exhibits intrinsic resistance to this drug.

(*) HECK-Yes bacteria

Intensive Care Units
University of Missouri – University Hospital
Units Include: CICU, MICU3, MICU5, NSICU, SICU
January 1, 2024-December 31, 2025

		# of Isolates	Penicillin (IV)	Oxacillin	Ampicillin	Ampicillin/sulbactam	Piperacillin/tazobactam	Cefazolin (urinary isolates only)	Cefoxitin	Cefotaxime/ceftriaxone	Ceftazidime	Cefepime	Meropenem	Amikacin	Gentamicin	Tobramycin	Clindamycin	Vancomycin	Tetracycline	Minocycline	Sulfamethoxazole/ trimethoprim	Ciprofloxacin	Levofloxacin	Nitrofurantoin (urinary isolates only)	# Isolates tested with nitrofurantoin
Gram Positive	<i>Staphylococcus aureus</i> (MSSA)	248	-	100	-	-	-	-	-	-	-	-	-	-	-	-	97	100	92	-	100	-	-	100	6
	<i>Staphylococcus aureus</i> (MRSA)	174	-	-	-	-	-	-	-	-	-	-	-	-	-	-	60	100	87	-	86	-	-	100	3
	<i>Staphylococcus coagulase negative</i> (CONS)	235	-	36	-	-	-	-	-	-	-	-	-	-	-	-	58	100	78	-	-	-	69	100	12
	<i>Streptococcus pneumoniae</i> [^]	77	100	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	75	-	100	-	-
	<i>Enterococcus</i> spp. (all)	127	-	-	67	-	-	-	-	-	-	-	-	-	-	-	72	-	-	-	-	-	-	80	56
	<i>Enterococcus faecalis</i>	79	-	-	100	-	-	-	-	-	-	-	-	-	-	-	97	-	-	-	-	-	-	100	40
	<i>Enterococcus faecium</i>	36	-	-	22	-	-	-	-	-	-	-	-	-	-	-	44	-	-	-	-	-	-	11	9
Gram Negative	<i>Enterobacter cloacae</i> [*]	63	-	-	-	-	-	0	-	-	-	92	100	-	100	-	-	-	-	-	90	98	93	43	7
	<i>Escherichia coli</i>	253	-	-	50	63	92	94	83	88	93	96	100	-	90	90	-	-	-	-	78	70	65	98	145
	<i>Klebsiella aerogenes</i> [*]	39	-	-	-	-	-	-	-	-	-	95	100	-	100	-	-	-	-	-	100	95	97	9	11
	<i>Klebsiella pneumoniae</i>	121	-	-	-	71	87	97	97	84	95	93	97	-	95	95	-	-	-	-	79	82	78	27	51
	<i>Proteus mirabilis</i>	52	-	-	76	81	100	-	94	90	-	96	100	-	83	-	-	-	-	-	69	69	74	-	-
	<i>Pseudomonas aeruginosa</i>	184	-	-	-	92	-	-	-	-	90	93	88	100	-	-	-	-	-	-	-	84	81	-	-
	<i>Serratia marcescens</i>	53	-	-	-	-	-	-	96	-	-	98	100	-	100	-	-	-	-	-	-	98	98	-	-
	<i>Stenotrophomonas maltophilia</i>	52	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	96	87	-	90	-	-

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Susceptibility is reported based on the number of isolates (at minimum 30 isolates) tested against a specific antimicrobial agent. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalexin. Data for fewer than 30 isolates per organism, per selected time frame is not statistically significant according to CLSI standards. Please exercise discretion when data are reviewed for species with fewer than 30 isolates. Gentamicin is no longer recommended for the management of *Pseudomonas* spp. per the Clinical and Laboratory Standards Institute.

Streptococcus pneumoniae isolates were identified from non-CSF cultures only.

(-) Drug was not tested, drug was not tested on >= 30 isolates, or drug is not indicated.

■ Organism exhibits intrinsic resistance to this drug.

(*) HECK-Yes bacteria

Fungal Organisms % Susceptible
University of Missouri – University Hospital
All locations and ages (excluding clinics)
January 1, 2024-December 31, 2025

	# of Isolates	Fluconazole	Voriconazole	Caspofungin	Micafungin
<i>Candida albicans</i>	42	100	100	100	100
<i>Candida dubliniensis</i>	6	100	100	-	-
<i>Nakaseomyces glabratus</i>	7	-	-	-	86
<i>Pichia kudriavzevii</i>	1		100	100	100
<i>Candida parapsilosis</i>	15	100	100	100	100
<i>Candida tropicalis</i>	8	100	100	100	100

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Data for fewer than 30 isolates per organism, per selected time frame is not statistically significant according to CLSI standards. Yeast antifungal susceptibilities are only performed per microbiology's policy (i.e. sterile body fluids, physician requests, etc.). Echinocandins are NOT appropriate for use in CNS and urinary infections. Taxonomy changes include *N. glabratus* (formerly *C. glabrata*) and *P. kudriavzevii* (formerly *C. krusei*).

(-) Drug was not tested or drug is not indicated.

▪ Organism exhibits intrinsic resistance to this drug.

Children's Hospital
University of Missouri – University Hospital
Units Include: NICU, Well Nursery, Peds, PICU, Youth Psych (excluding clinics)
January 1, 2024-December 31, 2025

		# of Isolates	Penicillin	Oxacillin	Ampicillin	Ampicillin/sulbactam	Piperacillin/tazobactam	Cefazolin (urinary isolates only)	Cefoxitin	Cefotaxime/ceftriaxone	Ceftazidime	Cefepime	Meropenem	Gentamicin	Tobramycin	Clindamycin	Erythromycin	Vancomycin	Rifampin^	Tetracycline	Minocycline	Sulfamethoxazole/ trimethoprim	Ciprofloxacin	Levofloxacin	Nitrofurantoin (urinary isolates only)	# Isolates tested with nitrofurantoin
Gram Positive	<i>Staphylococcus aureus</i> (MSSA)	60	-	100	-	-	-	-	-	-	-	-	-	-	-	96	-	100	-	95	-	100	-	-	-	-
	<i>Staphylococcus aureus</i> (MRSA)	45	-	-	-	-	-	-	-	-	-	-	-	-	-	91	-	100	100	93	-	76	-	-	-	-
	<i>Staphylococcus coagulase negative</i> (CONS)	65	-	21	-	-	-	-	-	-	-	-	-	-	-	33	13	100	-	94	-	-	-	44	-	-
	<i>Streptococcus pyogenes</i> (GAS)	11	100	-	-	-	-	-	-	-	-	-	-	-	-	100	100	100	-	-	-	-	-	-	-	-
	<i>Enterococcus</i> spp. (all species)	32	-	-	97	-	-	-	-	-	-	-	-	-	-	-	-	94	-	-	-	-	-	-	-	-
	<i>Enterococcus faecalis</i>	28	-	-	100	-	-	-	-	-	-	-	-	-	-	-	-	100	-	-	-	-	-	-	-	-
Gram Negative	<i>Enterobacter cloacae</i> *	15	-	-	-	-	-	-	-	-	-	100	-	100	-	-	-	-	-	-	-	100	100	100	-	-
	<i>Escherichia coli</i>	58	-	-	45	57	96	95	100	96	94	96	100	90	97	-	-	-	-	-	-	79	84	74	100	33
	<i>Klebsiella oxytoca</i>	12	-	-	-	75	-	-	100	92	-	100	100	92	-	-	-	-	-	-	-	92	92	91	-	-
	<i>Klebsiella pneumoniae</i>	21	-	-	-	100	100	-	100	100	-	100	100	100	-	-	-	-	-	-	-	95	100	100	-	-
	<i>Pseudomonas aeruginosa</i>	26	-	-	-	-	96	-	-	-	92	88	92	-	-	-	-	-	-	-	-	-	100	88	-	-
	<i>Raoultella</i> spp.	13	-	-	-	62	-	-	-	100	-	100	100	100	-	-	-	-	-	-	-	82	100	-	-	-
	<i>Stenotrophomonas maltophilia</i>	11	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	91	100	-	82	-	-

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Susceptibility is reported based on the number of isolates (at minimum **10** isolates) tested against a specific antimicrobial agent. Data for fewer than 30 isolates per organism, per selected time frame is not statistically significant according to CLSI standards. Please exercise discretion when data are reviewed for species with fewer than 30 isolates. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalexin. Gentamicin is no longer recommended for the management of *Pseudomonas* spp. per the Clinical and Laboratory Standards Institute.

(-) Drug was not tested or drug is not indicated.

^Rifampin is 100% susceptible against all *Staphylococcus aureus* strains. **Use only as adjunct to main therapy.**

▪ Organism exhibits intrinsic resistance to this drug.

(*) HECK-Yes bacteria

Cumulative Antimicrobial Resistance Report

University of Missouri – University Hospital

January 1-December 31, 2025

ESBL Rates, n/N (%)

	All Locations & Ages	Emergency Department	Intensive Care Units	Children's Hospital
<i>E. coli</i>	104/1265 (8)	104/1766 (6)	23/254 (9)	3/60 (5)
<i>K. oxytoca</i>	3/52 (6)	2/46 (4)	-	1/12 (8)
<i>K. pneumoniae</i>	45/322 (14)	36/352 (10)	12/122 (10)	0/21 (0)
<i>P. mirabilis</i>	10/176 (6)	10/211 (5)	3/53 (6)	-

Note: Emergency Department, Intensive Care Unit, and the Children's Hospital data are from 2024-2025

CRE Rates, n/N (%)

	All Locations & Ages	Emergency Department	Intensive Care Units	Children's Hospital
<i>E. coli</i>	4/1265 (0.3)	2/1766 (0.1)	1/254 (0.4)	0/60 (0)
<i>K. oxytoca</i>	0/57 (0)	0/48 (0)	-	0/12 (0)
<i>K. pneumoniae</i>	5/322 (1.6)	2/352 (0.6)	3/122 (2.5)	0/21 (0)
<i>P. mirabilis</i>	0/176 (0)	0/211 (0)	0/53 (0)	-
<i>A. baumannii</i>	15/27 (56)	-	-	-

Note: Based on CDC definition of CRE (i.e., Enterobacterales resistant to any carbapenem); Emergency Department, Intensive Care Unit, and the Children's Hospital data are from 2024-2025; *Acinetobacter baumannii* carbapenem resistance is reported for 2024-2025 for the "All Locations & Ages" category

Streptococcus pneumoniae Breakpoints (ug/mL)

Site	Penicillin	Cefotaxime/Ceftriaxone
Meningitis/CSF	IV: ≤ 0.06 (S); ≥ 0.12 (R)	≤ 0.5 (S); 1 (I); ≥ 2 (R)
Sterile, Non-CSF	IV: ≤ 2 (S); 4 (I); ≥ 8 (R)	≤ 1 (S); 2 (I); ≥ 4 (R)
Non-Sterile	PO: ≤ 0.06 (S); 0.12-1 (I); ≥ 2 (R)	Same as sterile, non-meningitis/CSF

Note: Susceptible (S); intermediate (I); resistant (R)

How to Perform Daily Antimicrobial Stewardship Reviews

Is a bacterial infection the most likely cause of the patient's symptoms?
If no, can antibiotics be stopped?

Is the narrowest spectrum antibiotic being used? If patient is on a fluoroquinolone, is a safer alternative available?

Can route be changed from IV to PO?

Is the antibiotic prescribed for the minimum effective duration?

Did you document dose, route, indication, and total planned duration in progress notes/discharge summary?

SHORTER IS BETTER (Get Stop-Dates)

Disease Process	Duration of Antibiotic
Community-acquired pneumonia	3-5 days
Hospital/ventilator pneumonia	5-7 days (depending on the pathogen isolated)
Cystitis	3-7 days (depending on agent)
Pyelonephritis/complicated urinary tract infection	5-7 days (depending on agent)
Catheter-associated urinary tract infection	3-7 days (depending on patient specific factors)
Skin & soft tissue infection	5-7 days
Non-purulent cellulitis	5 days
Intra-abdominal infection with source control	4 days
<i>Clostridioides difficile</i>	10 days
COPD exacerbation (when indicated)	5 days (3 days if azithromycin)
Sinusitis (when indicated)	5 days
Bacteremia	Uncomplicated MSSA/MRSA: 14 days Complicated MSSA/MRSA: 4-6 weeks Uncomplicated Gram-negative: 7 days Other uncomplicated: 7-14 days

Available from: <https://www.bradspellberg.com/shorter-is-better>

Blood Culture Rapid Diagnostics & Empiric Treatment

Gram-Positive Culture or BioFire Result	Suggested Initial Option	Notes
Coagulase-negative staphylococci (CoNS)	Requires clinical correlation (Often Contaminant) – Vancomycin initially to treat all isolates	These are all coagulase-negative <i>Staphylococci</i> (CoNS). When growing in only one set of blood cultures, this may represent contamination from the skin; however, clinical correlation is required. They are also often oxacillin resistant, not to be confused with oxacillin-resistant <i>S. aureus</i> . Consider stopping antimicrobials directed at this organism if infection is unlikely.
Methicillin-Susceptible <i>Staphylococcus epidermidis</i>		
Methicillin-Resistant <i>Staphylococcus epidermidis</i>		
Oxacillin-Susceptible <i>Staphylococcus lugdunensis</i>	Cefazolin or Nafcillin	Although this is a CoNS, it is considered a pathogen
Oxacillin-Resistant <i>Staphylococcus lugdunensis</i>	Vancomycin	
Oxacillin-Susceptible <i>Staphylococcus aureus</i>	Cefazolin or Nafcillin	ID consult recommended. <i>Staphylococcus aureus</i> is never a contaminant, and minimum duration of therapy is 14 days. Consult ID for appropriate further diagnostic approach and follow-up for extended IV/PO antimicrobials therapy.
Oxacillin-Resistant <i>Staphylococcus aureus</i>	Vancomycin	
<i>Streptococcus pneumoniae</i>	Ceftriaxone +/- Vancomycin*	*If concern for meningitis or severe infection add vancomycin until susceptibilities are known.
<i>Streptococcus pyogenes</i> (Group A)	Ampicillin or Cefazolin	Beta-hemolytic <i>Streptococci</i> , including <i>S. anginosus</i> , <i>S. agalactiae</i> , and <i>S. pyogenes</i> , are routinely susceptible to beta-lactams.
<i>Streptococcus agalactiae</i> (Group B)		
<i>S. anginosus</i> group		
<i>Streptococcus</i> species other than <i>S. pneumoniae</i> , <i>S. anginosus</i> group, <i>S. pyogenes</i> and <i>S. agalactiae</i>	Requires clinical correlation (Often Contaminant) – Ceftriaxone initially to treat	When growing in only one set of blood cultures, this may represent contamination from the skin; however, clinical correlation is required.
<i>Enterococcus faecalis</i> (non-VRE)	Ampicillin	ID Consult Recommended. If vanA or vanB are NOT detected vancomycin will likely be susceptible. Consult ID for appropriate further diagnostic approach and follow-up for extended IV/PO antimicrobial therapy.
<i>Enterococcus faecium</i> (non-VRE)	Vancomycin	
Vancomycin-resistant <i>Enterococcus faecalis</i> or Vancomycin-resistant <i>Enterococcus faecium</i>	Daptomycin or Linezolid	ID Consult Recommended. Concern for meningitis.
<i>Listeria monocytogenes</i>	Ampicillin	

Blood Culture Rapid Diagnostics & Empiric Treatment

Yeast Positive Culture or BioFire Result	Suggested Initial Option	Notes
<i>Candida albicans</i>	Fluconazole	Most commonly seen <i>Candida</i> species and is susceptible to fluconazole 100% of cases.
<i>Candida auris</i>	Micafungin	<i>C. auris</i> is commonly resistant to antifungals. Patient should be placed in isolation. ID consult recommended.
<i>Nakaseomyces glabratus</i> (formerly <i>Candida glabrata</i>)	Micafungin	De-escalation to fluconazole after susceptibilities are available is appropriate.
<i>Pichia kudriavzevii</i> (formerly <i>Candida krusei</i>)	Micafungin	<i>C. krusei</i> is routinely resistant to fluconazole.
<i>Candida parapsilosis</i>	Micafungin	<i>C. parapsilosis</i> is susceptible to micafungin 100% of cases.
<i>Candida tropicalis</i>	Micafungin or Fluconazole	Fluconazole susceptibilities vary. Micafungin recommended if patient is critically ill or unstable.
<i>Cryptococcus neoformans/gatti</i>	Amphotericin B	ID consult strongly recommended.
Detected as Yeast with no identification on BioFire	Micafungin	If yeast is present in the blood, even if not identified by BioFire, it is recommended to treat initially and consult ID for further management.

Consider ID Consult on all fungal species detected in the blood.

Blood Culture Rapid Diagnostics & Empiric Treatment

Gram-Negative Culture or BioFire Result	Suggested Initial Option		Notes
<i>Escherichia coli</i>	Continue current therapy with activity against gram negative bacteria		When ESBL is NOT detected, ceftriaxone, cefepime, and piperacillin-tazobactam are likely to cover.
<i>Klebsiella oxytoca</i>			
<i>Klebsiella pneumoniae</i>			
<i>Proteus spp.</i>			
<i>Salmonella spp.</i>	Ceftriaxone or Ampicillin-Sulbactam		Uncommon, but likely pathogenic
<i>Haemophilus influenzae</i>	Ceftriaxone or Ampicillin-Sulbactam		Beta-lactamase production is narrow. NOT ESBL.
<i>Neisseria meningitides</i>	Ceftriaxone		
<i>Enterobacter cloacae complex</i>	Cefepime		May harbor inducible AmpC beta-lactamase. Avoid ceftriaxone and piperacillin-tazobactam, even if micro report says susceptible unless discussed with ID. Cefepime is first line choice, unless cefepime MIC ≥ 4, then a carbapenem should be used.
<i>Klebsiella aerogenes</i>			
<i>Citrobacter freundii</i>			
<i>Hafnia alvei</i>			
<i>Yersinia enterocolitica</i>			
<i>Pseudomonas aeruginosa</i>	Piperacillin-Tazobactam or Cefepime		If patient status not improving or suspicion for multi-drug resistance, consider addition of tobramycin. ID consult recommended.
<i>Acinetobacter spp</i>	Ampicillin-Sulbactam (high dose)		Consult ID.
<i>Serratia marcescens</i>	Cefepime		Low risk of inducible AmpC production. If cefoxitin is resistant, recommend cefepime.
<i>Stenotrophomonas maltophilia</i>	Sulfamethoxazole-Trimethoprim		Minocycline has inadequate blood concentrations – don’t use if bacteremia. Requires clinical correlation (Often Colonizer).
<i>Bacteroides fragilis</i>	Include anaerobic coverage		Options that include good gram-negative anaerobic coverage include metronidazole, piperacillin-tazobactam, or ampicillin-sulbactam.
Gram-negative resistance markers may be present in above organisms	CTX-M (ESBL)	Ertapenem	Indicates ESBL producing organism, usually seen in <i>Klebsiella spp. and Escherichia coli</i> .
	KPC	Meropenem-Vaborbactam	ID consult required. Indicates a carbapenemase producer (CRE).
	OXA	Ceftazidime-Avibactam	ID consult required. When identified in <i>Acinetobacter spp.</i> Sulbactam-ampicillin is still the best empiric option, meropenem will likely be resistant.
	IMP or NDM or VIM	Cefiderocol	ID consult required. Indicates a CRE <u>not</u> susceptible to ceftazidime-avibactam.

Allergic Cross-Reactivity Potential Between Beta-Lactams

	Identical R1/R2 side chains – highest risk of cross-reactivity	Similar R1/R2 side chains – low risk of cross-reactivity but caution if history of life-threatening reaction
Ampicillin*	CEPHALEXIN, CEFACLOR	Cefadroxil, cefprozil
Amoxicillin*	CEFADROXIL, CEFPROZIL	Cephalexin, cefaclor
Cefaclor	AMPICILLIN, CEPHALEXIN	Penicillin, piperacillin, amoxicillin, cefadroxil, cefprozil
Cefadroxil	AMOXICILLIN, CEFPROZIL	Penicillin, piperacillin, ampicillin, cephalexin, cefaclor
Cefazolin	NONE	NONE
Cefdinir	CEFIXIME	Ceftibuten, ceftriaxone, cefotaxime, cefpodoxime, ceftazidime, cefepime, ceftaroline, ceftolozane, cefiderocol
Cefepime	CEFTRIAZONE, CEFOTAXIME, CEFPODOXIME	Cefuroxime, ceftibuten, cefdinir, cefixime, ceftazidime, ceftaroline, ceftolozane, cefiderocol
Cefixime	CEFDINIR	Cefuroxime, ceftibuten, ceftriaxone, cefotaxime, cefpodoxime, ceftazidime, cefepime, ceftaroline, ceftolozane, cefiderocol
Cefoxitin	CEFUROXIME	Cefotaxime
Cefpodoxime	CEFTRIAZONE, CEFEPIME, CEFOTAXIME	Cefuroxime, ceftibuten, cefdinir, cefixime, ceftazidime, ceftaroline, ceftolozane, cefiderocol
Cefprozil	AMOXICILLIN, CEFADROXIL	Penicillin, piperacillin, ampicillin, cephalexin, cefaclor
Ceftaroline	NONE	Cefuroxime, ceftibuten, cefdinir, cefixime, ceftriaxone, cefotaxime, cefpodoxime, ceftazidime, cefepime, ceftolozane, cefiderocol
Ceftazidime	AZTREONAM, CEFIDEROCOL	Ceftibuten, cefdinir, cefixime, ceftriaxone, cefotaxime, cefpodoxime, cefepime, ceftaroline, Ceftolozane
Ceftibuten	NONE	Cefdinir, cefixime, ceftriaxone, cefotaxime, cefpodoxime, ceftazidime, cefepime, ceftaroline, ceftolozane, cefiderocol
Ceftolozane	NONE	Ceftibuten, cefdinir, cefixime, ceftriaxone, cefotaxime, cefpodoxime, ceftazidime, cefepime, ceftaroline, cefiderocol
Ceftriaxone	CEFEPIME, CEFOTAXIME, CEFPODOXIME	Cefuroxime, ceftibuten, cefdinir, cefixime, ceftazidime, ceftaroline, ceftolozane, cefiderocol
Cefuroxime	CEFOXITIN	Cefixime, ceftriaxone, cefotaxime, cefpodoxime, cefepime, ceftaroline
Cephalexin	AMPICILLIN, CEFACLOR	Penicillin, piperacillin, amoxicillin, cefadroxil, cefprozil
Penicillin G*		Cefadroxil, cephalexin, cefprozil, cefaclor
*Chart addresses cross-reactivity with cephalosporins. Avoid penicillins in true penicillin allergy.		

Allergic Cross-Reactivity Potential Between Beta-lactams

<i>β</i>-Lactam Antibiotic	Amoxicillin	Penicillin G	Penicillin V	Flucloxacillin	Feneticillin	Piperacillin	Cephalexin	Cefazolin	Cefalothin	Cefuroxime	Cefaclor	Cefamandol	Ceftibuten	Ceftriaxone	Cefotaxime	Ceftazidime	Cefepime	Cefiderocol	Ceftaroline	Ceftolozane	Meropenem	Imipenem	Ertapenem	Aztreonam
Amoxicillin	✓							✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Penicillin G		✓						✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Penicillin V			✓					✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Flucloxacillin				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Feneticillin					✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Piperacillin						✓		✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cephalexin							✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cefazolin	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cefalothin	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cefuroxime	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓								✓	✓	✓	✓
Cefaclor				✓	✓			✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cefamandole				✓	✓			✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ceftibuten	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓									✓	✓	✓	✓
Ceftriaxone	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓									✓	✓	✓	✓
Cefotaxime	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓									✓	✓	✓	✓
Ceftazidime	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓									✓	✓	✓	✓
Cefepime	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓									✓	✓	✓	✓
Cefiderocol	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓									✓	✓	✓	✓
Ceftaroline	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓									✓	✓	✓	✓
Ceftolozane	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓									✓	✓	✓	✓
Meropenem	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				✓
Imipenem	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				✓
Ertapenem	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				✓
Aztreonam	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓									✓	✓	✓	✓

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	Cross-tabulation was similar
	Allergy possible based on the formation of PPL
	Potential cross-allergy based on an identical R1 side chain
	Potential cross-allergy based on similarity in R1 or R2 side chains or clinical studies
✓	No risk of a cross-allergic reaction

Last updated: March 7, 2026